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Soundless: A study on noise sleep disturbance

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Abstract

The Soundless [1] project carried out by the Universitat Rovira i Virgili (Tarragona, Spain) aims to be a citizen science platform to study how noise affect rest in different neighbourhoods of the city of Tarragona.

The soundless project gathered 40 neighbours of the city of Tarragona living in the neighbourhoods of El Serrallo and Barri del port and conducted a study on how the sleep is disturbed by train lines noise. Tarragona is a port city located in north-east Spain (Catalonia) by the Mediterranean Sea. A major problem is that railway tracks create a barrier between the city and the sea (beach, port). Apartment blocks are located just at 2-5 meters of distance from the tracks without any acoustic barriers.

The first challenge of the Soundless project is to gather the data to conduct an study on sleep quality. To do so they used a FitBit (an intelligent bracelet) able to capture the vital signs of an individual, such as the heart rate, the oxygen saturation, etc. Once the data is gathered the next step is to detect an incidence in a reliable manner, here we define a sleep incident as when a person moves from a deep sleep phase to being completely awake due to excessive noise. To do so several algorithms are applied to the data in order to detect anomalies in sleep related to noise in the ambient. A first approach was to use direct mathematical correlation between heart rate and noise, some factors affected this measurement however:

- Individual Variability: Participants exhibited diverse heart rate patterns and sensitivities to noise.
- Data Complexity: Recordings contained inherent noise even after filtering, adding to the challenge of isolating the influence of sound incidents.
- Reaction Time Differences: Individual response times to noise varied considerably, ranging from 10 to 40 seconds, further complicating correlation analysis.
- Sensitivity Fluctuations: Individual sensitivity to noise appeared to fluctuate, with inconsistent reactions even within the same participant

- Sleep Stage Influences: Natural heart rate variations during different sleep stages added another layer of complexity.

What led the Soundless team to adopt z-scores algorithm. With this approach they were successfully able to detect incidents in heart rate and in sound measurements, and, afterwards relate the incidents in heart rate to incidents in sound. Their results show that the legal limit set by the World Health Organization (WHO) of 45 decibels was exceeded on 98% of nights with recordings. Additionally they show that each and every user has, in average, 12 sound incidents during the night and half of the nights that a recording was taken an incident in their heart rate is detected. Additionally one in five nights each user has been woken up due to a high level of ambient noise.

Soundless study carry on trying to detect individuals with a higher sensitivity to noise in the ambient, having individuals with values close to 38 dB to have a disturbance and others that requires louder noises. Particularly, they show that 70.59% of the times a sleep disturbance has been detected, it has occurred in people with a higher sensitivity to noise, in this case, a sensitivity less than 45 decibels. On the other hand, people with lower sensitivity to this type of situation have only registered 29.41% of the sleep disturbances caused by noise. After the study on individual persons they examine general sensitivity to noise in the ambient, where they find that values between 40 and 50 dB are able to create incidences in the heart rate and in the sleep process of almost any individual.

The study concludes that the night noise produced by freight trains in the seaside of Tarragona is far beyond legal limits ($>45\text{dB}$) and up to 80dB during all nights. Furthermore, they consistently detected the impact of noise in health variables like heart rates and sleep modes, which affected most users in the study during all nights. These disturbances during sleep are causing irregular heart rates, sudden awakenings, and noticeable impacts in sleep quality in the citizens.

Introduction

Sleep quality is a crucial factor in maintaining overall health. In today's world, where a significant portion of the population resides in densely populated urban areas (megapolises), people are continuously exposed to noise pollution. This constant exposure to ambient noise day and night raises concerns about its potential impact on sleep. Investigating how environmental noise affects sleep quality is therefore a highly relevant topic. In this document, we present a study focused on this issue, primarily through the analysis of existing datasets.

We begin by working with the Human Sleep Project dataset [2], developed by Massachusetts General Hospital, which includes around 25,000 electroencephalogram (EEG) recordings from subjects with various sleep pathologies. These recordings, stored in the .edf (European Data Format), often come with detailed annotations and information on the EEG channels used. The dataset has been widely studied by researchers across multiple disciplines. Leveraging this resource, we aim to develop an artificial intelligence-based solution capable of automatically annotating EEG recordings to assist in sleep quality analysis.

Next, we introduce the Soundless [1] dataset, part of a research project led by Universitat Rovira i Virgili in Tarragona, Spain. This project is designed to study the impact of environmental noise on sleep quality through the use of FitBit devices, which monitor participants' vital signs during sleep, and a companion Android application that records ambient sound throughout the night. While the dataset currently includes data from approximately 40 participants, it provides valuable real-world insight into how different noise conditions may influence various stages of sleep. It also acknowledges that individuals may have different sensitivities to noise, and that certain noise frequencies might affect sleep differently across subjects.

The objective of this work is to bridge the controlled, large-scale data of the Human Sleep Project with the real-world, noise-focused observations of the Soundless dataset. By connecting these two sources, we aim to extrapolate key metrics and draw meaningful conclusions about the relationship between environmental noise and sleep quality.

The following pages present an initial approach to studying sleep quality by combining data-driven analysis with real-world noise exposure scenarios. The primary aim of this work is to explore how different stages of sleep are influenced by environmental noise, using a combination of two complementary datasets: the Human Sleep Project and the Soundless dataset. Through this exploratory study, we aim to build a foundation for further research into sleep quality assessment in urban and noise-rich environments.

The main goals of this document are as follows:

- Design a method for analysing sleep stages in subjects from the Human Sleep Project dataset. This includes processing and interpreting electroencephalogram (EEG) signals to identify key sleep phases such as REM, light sleep, and deep sleep, with a focus on creating a robust and scalable analysis pipeline.
- Define a set of metrics derived from the identified sleep stages to establish a meaningful connection between participants in the Soundless dataset and those in the Human Sleep Project. These metrics may include sleep stage duration, frequency of transitions between stages, interruptions, and overall sleep efficiency, among others.
- Draw data-driven conclusions about the quality of sleep in environments with varying levels of noise. By correlating ambient sound data with changes in sleep patterns, we aim to understand how specific noise characteristics (e.g., intensity, frequency, duration) may affect different individuals and sleep stages, ultimately influencing overall sleep quality.

This preliminary study sets the groundwork for future research that could lead to better tools for monitoring and improving sleep in urban settings, potentially informing both clinical practice and consumer health applications.

State of art

The relationship between environmental noise and sleep quality has been the focus of extensive interdisciplinary research due to its profound implications for public health. Numerous studies have demonstrated that nocturnal noise exposure adversely affects both the physiological and psychological dimensions of sleep, including sleep architecture, duration, and subjective sleep quality.

One of the most influential sources on this topic is the World Health Organization's environmental noise guidelines, which synthesize extensive epidemiological and experimental evidence. Basner and McGuire (2018), in their systematic review, concluded that night time exposure to noise from traffic, aircraft, and industrial sources leads to delayed sleep onset, increased number and duration of awakenings, and reduced deep sleep (slow-wave sleep). These findings support earlier work by Muzet (2007), who outlined the biological pathways through which noise interferes with sleep, such as through increased heart rate, blood pressure, and cortical arousals.

Different types of environmental noise have been shown to affect sleep in varying degrees. Griefahn et al. (2006) and Pirrera, De Valck, and Cluydts (2010) reviewed the impacts of road, rail, and air traffic noise, noting that aircraft noise tends to be the most disruptive due to its variability and unpredictability. Basner et al. (2008) used polysomnography to demonstrate that aircraft noise exposure reduces the proportion of REM and slow-wave sleep, both critical for physical recovery and cognitive function.

Recent studies have also investigated the role of individual differences in noise sensitivity. Elmenhorst et al. (2012) found that individuals with higher subjective sensitivity to noise are more prone to awakenings and have lower sleep efficiency under noisy conditions. Similarly, Riedel et al. (2022) showed that noise annoyance significantly moderates the relationship between traffic noise exposure and sleep quality, indicating that psychological factors mediate the perception and physiological impact of noise.

Urbanization and socio-economic status are additional contextual factors that modulate the effects of noise. Dzhambov et al. (2017), through meta-analysis, found that people living in urban and low-income neighbourhoods are disproportionately affected by road

traffic noise, suggesting environmental justice implications. Brown and van Kamp (2009) emphasized the role of subjective appraisal and context in shaping noise response, indicating that perceptions of control and expectations may alter physiological responses to the same noise level.

The long-term consequences of chronic noise-induced sleep disruption are also well-documented. Witte et al. (2021) linked prolonged night-time noise exposure to cognitive decline in older adults, while Dratva et al. (2012) showed that noise-related sleep disturbance is associated with reduced health-related quality of life and increased psychological distress.

In sum, the current body of research provides robust evidence that environmental noise impairs sleep quality through both direct physiological mechanisms and mediated psychological pathways. While regulation and mitigation strategies have been proposed, further research is needed to explore personalized interventions and to better understand vulnerable populations.

To the best of our knowledge, no prior research has established a relationship between the Human Sleep Project and the Soundless dataset. This study introduces a novel methodological approach aimed at bridging these two datasets, with the objective of advancing our understanding of how environmental noise influences sleep quality.

Analysis

A critical step in this study involves understanding and preprocessing the data from the Human Sleep Project. Given the scale and complexity of this dataset, careful examination is required before any meaningful analysis can be performed.

EEG montages are organized arrangements of channels, where each channel represents the potential difference between two electrodes, used to display and interpret scalp electrical activity. They are crucial for visualizing brain activity, enabling the appreciation of waveforms from different perspectives, aiding in lateralization and localization of potential sources, and facilitating artifact identification.

To study the Human Sleep Project dataset, the first task is to analyze its vast and highly heterogeneous structure. The dataset consists of over 25,000 electroencephalogram (EEG) recordings, each captured using different channel configurations. As a starting point, it is essential to understand the various EEG montages used across recordings and to develop a unified framework that allows consistent analysis across the dataset. Within the collection, there are 23 distinct montage configurations, each comprising the same number of channels but often differing in the specific channels included. To address this variability, I conducted an analysis of channel frequency across all configurations, aiming to identify a common subset that would enable a standardized and comprehensive study of the data.

The study yields the following results:

Channel	Frequency (%)
ABD	88.719512
AIRFLOW	75.304878
C3-M2	71.951220
CHEST	87.042683
E1-M2	84.451220
EKG	87.957317
HR	89.786585
IC	89.329268
LAT	82.164634
O1-M2	70.731707
RAT	79.878049
SNORE	91.920732
SaO2	90.396341

Table 1: Channel frequencies

In table 1 we can see the most common channels in the dataset, and the associated frequency of the channel across all the electroencephalograms in the dataset. With this information we can impose that our electroencephalograms have, at least, this channels and we must get a representation of around 71% of the dataset.

Going one step further we can impose channel equivalences. We define a channel equivalence as a pair of channels that, if exchanged between configurations, we obtain the same configuration. Studying the different configurations present in the dataset we obtain the following equivalences:

Channel equivalences								
C4-M1	CHIN1-CHIN2	E1-M1	E2-M1	EKG	F4-M1	O2-M2	LAT	RAT
C4-M2	CHIN1-CHIN3	E1-M2	E2-M2	EKG-E1	F4-M2	O1-M1	LAT-E1	RAT-E1
	CHIN2-CHIN3				F8-M1	O2-M1		
	CHIN3-CHIN1				Fp2-M1			
	CHIN3-CHIN2				Fp2-M2			
	Chin1-31							
	Chin1-Chin2							
	Chin1-Chin3							
	Chin1-P3							
Chin1-P4								
Chin2-Chin3								
Chin3-Chin2								

Channel	Frequency (%)	Equivalence
ABD	88.719512	No
AIRFLOW	75.304878	No
C3-M2	71.951220	No
C4-M1	76.371951	Yes
CHEST	87.042683	No
CHIN1-CHIN2	91.310976	Yes
E1-M1	85.365854	Yes
E2-M1	87.347561	Yes
EKG	90.396341	Yes
F4-M1	81.402439	Yes
HR	89.786585	No
IC	89.329268	No
LAT	84.451220	Yes
O1-M2	70.731707	No
O2-M2	86.280488	Yes
RAT	82.164634	Yes
SNORE	91.920732	No
SaO2	90.396341	No

In table [Table 2](#) I summarize the frequencies obtained when aggregating the equivalences found by the code, after all we have 18 meaningful channels to study, however we need to take into account that most of them are not related to brain activity, so looking for brain activity on those channels will make no sense. However we can try to use the 18 channels to feed a neural network as they are, as a sequence of voltages per channel over time.

Power Spectral Density (PSD) analysis is a widely used quantitative EEG (qEEG) technique for examining the spectral composition of EEG signals during sleep. It provides valuable insights into various sleep stages and can aid in the identification of potential sleep disorders by measuring the distribution of power across different frequency bands within the EEG signal.

Delta activity

Delta brainwaves, also known as delta activity, are the slowest brainwaves, typically between 0.5 and 4.5 Hz. They are primarily associated with deep, dreamless sleep, where the body is in a state of complete relaxation and the mind is minimally active. Delta waves are also observed in newborns, and can be induced in a waking state by experienced meditators.

They usually show a high amplitude and are most prevalent during deep sleep stages (3), where the brain is essentially shut down for repair and regeneration. Delta waves are linked to various restorative processes, including tissue repair, muscle growth, and hormone release. They are also involved in transferring information from the hippocampus to the neocortex, aiding in long-term memory storage.

Theta activity

Theta brainwaves, characterized by frequencies between 4.5-8.5 Hz, are associated with states of relaxation, deep meditation, and creativity. They are often observed during sleep, particularly during dream sleep, and are also linked to states of daydreaming and focused attention. Theta activity is thought to play a role in processing information, making memories, and facilitating intuitive insights.

Disrupted or divergent theta activity has been linked to various mental health conditions.

Alpha activity

Alpha brainwaves, with a frequency of 8.5-11.5 Hz, are associated with a relaxed, awake state, often observed during activities like daydreaming or meditation. They indicate the brain is active but not overly focused, promoting feelings of calmness and ease.

Alpha activity is typically strongest over the posterior areas of the head, particularly the occipital cortex. Alpha waves can be influenced by various factors, including sensory stimuli, mental activities, and even the position of the eyes (they are often more prominent with eyes closed).

Sigma activity

Sigma activity, also known as sigma power or spindle activity, refers to brainwave activity in the frequency range of 11.5-15.5 Hz, a key component of non-REM sleep. These brainwaves are characterized by transient oscillations and are often associated with sleep-dependent memory consolidation and sleep stability. In infants, sigma activity is linked to psycho motor development and local brain maturation.

Beta activity

Beta activity in the brain refers to a range of brainwave frequencies, typically between 15.5 and 30 Hz, associated with alert, focused, and active mental states. These brainwaves are commonly present during waking hours and when engaged in tasks requiring concentration, problem-solving, or decision-making.

An excessive beta activity can be associated with symptoms of brain over-arousal, such as anxiety, obsessiveness, and sleep difficulties, while a deficient beta activity may be linked to difficulties concentrating, problem-solving, and overall brain under-arousal.

Motivated by findings from recent literature [3], we can apply PSD analysis to extract the frequency spectrum from the EEG recordings, resulting in a feature vector that quantifies the power observed in each channel across frequency bands associated with brain activity. However, upon closer examination, it became evident that the set of EEG channels initially considered was not optimal for detecting brain-related spectral features. Therefore, the list of channels used in the analysis was refined and reduced to a smaller, more representative subset. The selected channels used for PSD analysis are as follows:

Channel	Frequency (%)	Equivalence
C3-M2	71.951220	No
C4-M1	76.371951	Yes
E1-M1	85.365854	Yes
E2-M1	87.347561	Yes
F4-M1	81.402439	Yes
O1-M2	70.731707	No
O2-M2	86.280488	Yes

Table 3: Channels for PSD Analysis

Design

In order to establish a meaningful relationship between the Human Sleep Project dataset and the Soundless dataset, it is essential to standardize the analysis of the Human Sleep Project data. This can be achieved by predicting the sleep stage of a subject at each moment within their electroencephalogram (EEG) recording. Once sleep stages are accurately classified, relevant sleep metrics can be computed, enabling comparisons between the two datasets.

To perform this classification, a neural network presents itself as a suitable tool due to its capacity to model complex patterns in time-series data. The goal is to classify each segment of the EEG into one of the following sleep stages:

- **Sleep_stage_W:** This stage indicates that the subject is awake.
- **Sleep_stage_N1:** This stage indicates that the subject is in the first sleep stage.
- **Sleep_stage_N2:** This stage indicates that the subject is in the second sleep stage.
- **Sleep_stage_N3:** This stage indicates that the subject is in the third sleep stage.
- **Sleep_stage_R:** This stage indicates that the subject is in REM (Rapid Eye Movement) stage.

As stated in previous sections, the initial approach adopted was to analyze the electroencephalograms in their original form, treating them as sequences of voltage measurements recorded over time for each channel. To this end, I employed a fully convolutional neural network (FCN), which is particularly well-suited for modeling temporal dynamics and spatial relationships present within multi-channel EEG data.

The choice of an FCN was motivated by its ability to efficiently extract features from time-series data without relying on traditional recurrent architectures. This approach leverages convolutional filters to capture both short-term fluctuations and longer-term dependencies within the signal, thereby enabling a comprehensive analysis of the electroencephalogram's intrinsic structure. Furthermore, FCNs have been demonstrated to perform ro-

bustly in similar biomedical applications, making them an attractive option for tasks such as sleep stage classification and signal anomaly detection.

By processing the EEG recordings in their raw format, this method allows for an end-to-end feature extraction and classification framework, whereby the network learns to identify subtle patterns associated with different sleep stages directly from the voltage sequences. The following sections detail the network architecture, the training regimen, and the evaluation metrics used to assess the performance of this approach in analyzing and classifying the EEG data.

Fully Convolutional neural network (FCN)

Based on the paper Time Series Classification from Scratch with Deep Neural Networks: A Strong Baseline [4] and the related work found at [5] I build a fully convolutional network for time series classification, the architecture is as follows:

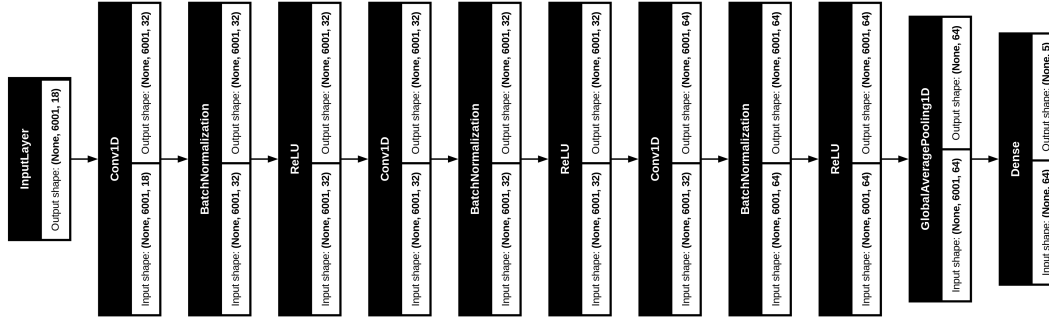


Figure 1: FCN architecture

The input shape to the fully convolutional neural network is (6001, 18), where 18 corresponds to the number of EEG channels and 6001 to the number of time points in each segment. Each input segment represents a 30-second window of the EEG signal, sampled at 200 Hz, resulting in 6000 samples, plus one additional point depending on the indexing used during preprocessing. For consistency, only EEG recordings sampled at 200 Hz are retained; any files with a different sampling rate are excluded from this approach.

The network architecture begins with a 1D convolutional layer consisting of 32 filters and a kernel size of 3, applied along the temporal axis. This is followed by a second convolutional layer, also with 32 filters, but using a kernel size of 5 to capture slightly broader temporal features. A third convolutional layer with 64 filters and a kernel size of 5 is then used to further increase the representational capacity of the network.

To reduce dimensionality and extract global temporal features, an average pooling layer

is applied after the convolutional blocks. Finally, the output is passed through a dense (fully connected) layer with 5 neurons, corresponding to the number of target sleep stages or classes in the classification task.

The activation function used for all hidden layers in the network is the Rectified Linear Unit (ReLU), chosen for its computational efficiency and its ability to mitigate the vanishing gradient problem during training. For the output layer, a softmax activation function is applied, as the task involves multi-class classification, with the network output representing a probability distribution over the defined sleep stage classes.

The hyperparameters of the neural network were optimized using Keras Tuner, a tool designed to automate the selection of optimal configurations for deep learning models. While Keras Tuner provides a structured and efficient way to explore hyperparameter spaces, its integration with this project posed certain limitations. Specifically, because the full dataset is large and hosted externally, it was not feasible to perform tuning across the entire dataset. As a result, the tuning process was conducted on a representative subset of the data. Despite this constraint, Keras Tuner proved to be a valuable aid in guiding the selection of key model parameters such as the number of filters, kernel sizes, and learning rates.

Several optimization strategies were evaluated to determine the most effective training configuration for the proposed network architecture. In particular, the Adam, Stochastic Gradient Descent (SGD), and AdamW optimizers were tested. For the AdamW optimizer, a weight decay parameter of $1 * 10^{-4}$ was applied to regularize the model and prevent overfitting.

Additionally, a range of learning rates between $1 * 10^{-3}$ and $1 * 10^{-5}$ was explored during training to assess their impact on convergence and performance. A learning rate scheduler was also employed to dynamically adjust the learning rate over time, further improving training stability and efficiency.

LSTM

Long Short-Term Memory (LSTM) networks [6] are a type of recurrent neural network (RNN) specifically designed to model and learn from sequential data. Unlike traditional RNNs, which suffer from vanishing or exploding gradients over long sequences, LSTMs incorporate memory cells and gating mechanisms that allow them to capture long-range dependencies and temporal dynamics more effectively. This makes them particularly well-suited for time series classification tasks, such as analyzing electroencephalogram (EEG) signals, where temporal patterns play a crucial role in identifying different sleep stages.

Given these strengths, an alternative model architecture was developed by replacing the

final two convolutional layers in the original network with one or more LSTM layers. The rationale behind this change is that while convolutional layers are effective at extracting local spatial and short-term temporal features, LSTMs may provide improved performance by explicitly modeling the temporal evolution of brain activity across the EEG signal. This hybrid approach leverages the strength of convolutional layers for feature extraction at the lower levels of the network, while utilizing LSTMs to capture higher-order temporal dependencies at later stages.

The architecture proposed is as follows:

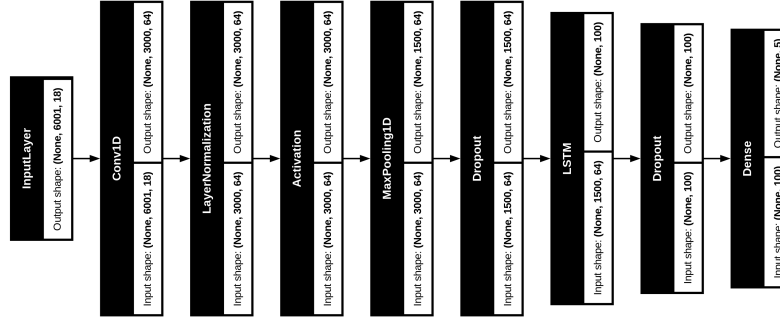


Figure 2: CN-LSTM architecture

The input layer remains unchanged, with a shape of (6001, 18), corresponding to 30-second EEG segments sampled at 200 Hz across 18 channels. The network begins with an initial 1D convolutional layer consisting of 64 filters and a kernel size of 3, which is effective for capturing local temporal patterns in the data. This is followed by a normalization layer, which helps stabilize the learning process, and a ReLU activation function to introduce non-linearity. A pooling layer is then applied to downsample the feature maps and reduce computational complexity.

To mitigate overfitting, a dropout layer is added next. This layer randomly deactivates a fraction of the neurons during training, helping to prevent the model from becoming too sensitive to noise in the input data and promoting generalization. Following this, a Long Short-Term Memory (LSTM) layer with 100 units is introduced, using the default tanh activation function, to capture the sequential dependencies in the EEG signal.

Another dropout layer is applied after the LSTM to further regularize the model. Finally, the architecture concludes with a dense output layer containing 5 neurons, corresponding to the number of sleep stage classes in the classification task. This output layer uses a softmax activation function to produce class probabilities.

For this architecture, several optimizers were evaluated, including Adam with learning rates ranging from $1 * 10^{-3}$ to $1 * 10^{-5}$, as well as AdamW, incorporating a weight decay

parameter of $1 * 10^{-4}$ to improve regularization.

Multi Layer Perceptron (MLP)

A Multi-Layer Perceptron (MLP) [7] is a type of feedforward neural network characterized by an input layer, one or more hidden layers, and an output layer. MLPs are particularly well-suited for classification tasks involving tabular or fixed-length feature data.

Following the Power Spectral Density (PSD) analysis, we obtain a fixed-length feature vector representing the power distribution across frequency bands related to brain activity for each selected EEG channel. Given this structured feature representation, a straightforward MLP architecture is an appropriate choice for classification. The specific architecture employed in this study is as follows:

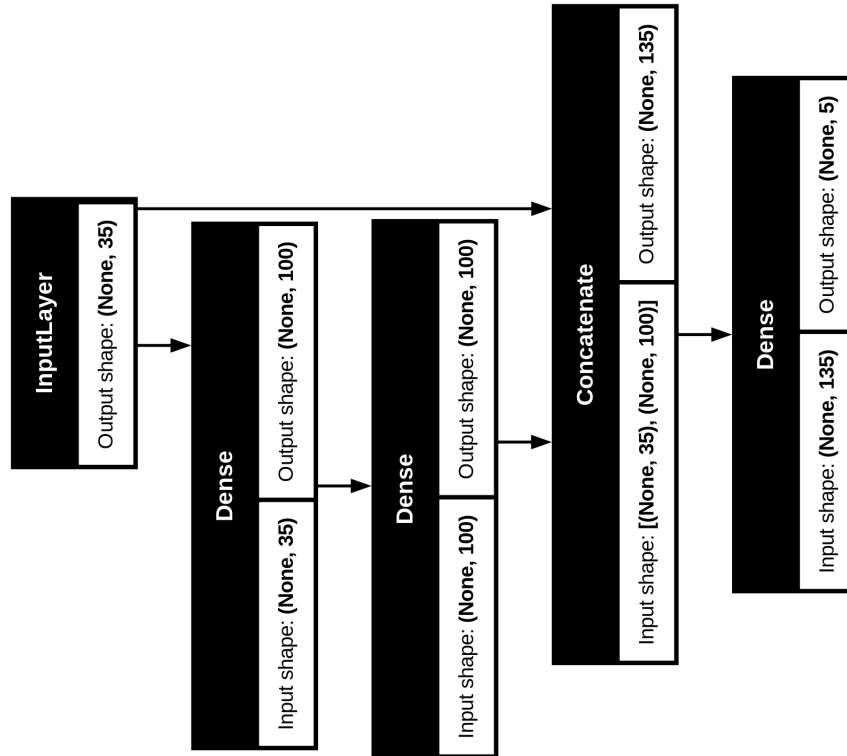


Figure 3: MLP architecture

The input layer has a shape of (35) , corresponding to the 7 selected EEG channels multiplied by the 5 frequency bands of interest derived from the PSD analysis. This input is first

passed through a dense layer with 100 neurons and a ReLU activation function, followed by a second hidden dense layer, also with 100 neurons and ReLU activation.

After these hidden layers, the original input vector is concatenated with the output of the second hidden layer, creating a model with both a deep path (through the hidden layers) and a wide path (direct input). This architecture leverages both the learned high-level features and the raw input features, which can enhance classification performance when the input features are strongly representative of the target classes.

Finally, the concatenated output is passed to a dense output layer with 5 neurons and a softmax activation function, corresponding to the five sleep stage classes to be predicted.

The optimizer selected for training the MLP is Adam, initialized with a learning rate of $1 * 10^{-3}$. To enhance training stability and promote efficient convergence, a ReduceLROnPlateau learning rate scheduler is employed. This scheduler reduces the learning rate automatically when the validation error plateaus, allowing the network to focus more effectively on minimizing the loss and achieve a more stable, albeit slower, convergence.

The hyper parameters, including the number of neurons and the learning rate for this network, were optimized using Keras Tuner. Multi-Layer Perceptrons are highly sensitive to data scaling, as they typically assume input features are centered around zero with a standard deviation of one. To meet this assumption, the data preprocessing pipeline first normalizes the samples to have zero mean and unit variance, followed by scaling each sample using a MinMaxScaler. This two-step scaling process significantly improved the stability of the network during training and led to a notable increase in classification accuracy.

Implementation

This section provides an overview of the technologies and tools used to implement the methodology described above. Given the size and complexity of the dataset, comprising over 25,000 files, each averaging approximately 500 MB, the neural network training process is non-trivial. It involves multiple stages: first, downloading the data; second, preprocessing and aggregating the data; and finally, training the neural network models.

The entire implementation is developed in Python, a widely-used programming language for artificial intelligence and machine learning tasks. Python's rich ecosystem of libraries, including Keras, TensorFlow, and scikit-learn, are leveraged extensively throughout this project.

For the data acquisition phase, the boto3 library is utilized to interface with AWS S3 storage services, from which the dataset files are downloaded.

Data preprocessing is facilitated by the MNE library, which is specifically designed for handling electrophysiological data formats such as .edf files. MNE enables efficient extraction of EEG epochs corresponding to specific annotations and the computation of Power Spectral Densities (PSDs) for each event.

To manage and organize the processed data, a PostgreSQL relational database is employed. Preprocessed data is aggregated into batches and stored within this database, allowing the neural network training routines to efficiently access and load data in manageable chunks.

Finally, the neural networks themselves are implemented using Keras, a high-level deep learning library that abstracts the complexities of model creation and training, enabling rapid prototyping and experimentation.

Evaluation

Among the three approaches presented in the Design chapter, namely the Fully Convolutional Network (FCN), the FCN combined with a Long Short-Term Memory (LSTM) layer, and the Multi-Layer Perceptron (MLP), only the MLP architecture produced meaningful results. Despite efforts to enhance performance, the FCN, even when augmented with an LSTM layer to better capture temporal dependencies, consistently yielded poor classification outcomes. Specifically, the best validation accuracy achieved by this hybrid architecture hovered around 45%, which, although better than random guessing in a five-class setting, remains insufficient for reliable sleep stage classification.

These results suggest that the convolutional and recurrent models struggled to generalize well on the available data, potentially due to factors such as high variability in EEG channel configurations, noise in the recordings, or insufficient temporal context learned by the network. In contrast, the MLP, trained on well-structured and aggregated frequency-domain features obtained through PSD analysis, demonstrated a more stable and effective learning process.

Given this outcome, the focus of this section will be on analyzing the results obtained using the MLP model. We will explore its classification performance in detail, including accuracy, confusion matrix, and other relevant evaluation metrics, to assess its effectiveness in predicting sleep stages from EEG-derived features.

The Multi-Layer Perceptron (MLP) model was trained on data from 1,437 subjects, each contributing an average of approximately 850 annotated events, resulting in a total of over 1.2 million events used during the training phase. The model achieved a validation accuracy of nearly 70%, indicating a strong ability to generalize to unseen data within the validation set.

To further evaluate the model's performance, a separate test set was constructed using data from 124 subjects randomly selected from the dataset. Importantly, these subjects were excluded from the training phase, ensuring that the test results reflect the model's ability to generalize to entirely new individuals. The test set comprised approximately 102,000 events, and serves as a robust benchmark for assessing the real-world applicabil-

ity of the trained MLP model.

The classification report obtained for the predictions computed with the resulting model is the following:

	precision	recall	f1-score	support
Sleep_stage_W	0.59	0.60	0.60	13843
Sleep_stage_N1	0.33	0.26	0.29	10978
Sleep_stage_N2	0.80	0.77	0.78	47168
Sleep_stage_N3	0.70	0.73	0.71	13242
Sleep_stage_R	0.65	0.78	0.71	17240
accuracy			0.69	102471
macro avg	0.61	0.63	0.62	102471
weighted avg	0.68	0.69	0.68	102471

Table 4: Classification report

As shown in the classification report, the model’s primary weakness lies in its performance on the Sleep_stage_N1 class, where it achieves a precision of only 33%. Upon closer examination, it becomes evident that this class also has the lowest support among all classes, indicating a significant class imbalance in the dataset. This imbalance likely hinders the model’s ability to learn distinguishing features for N1-stage events, resulting in poor generalization and increased misclassification.

To address this issue, the application of data augmentation techniques, such as SMOTE (Synthetic Minority Over-sampling Technique), could help balance the class distribution by generating synthetic examples of underrepresented classes. This, in turn, may improve the model’s sensitivity and predictive performance for Sleep_stage_N1 in future iterations.

The confusion matrix obtained during evaluation is shown below:

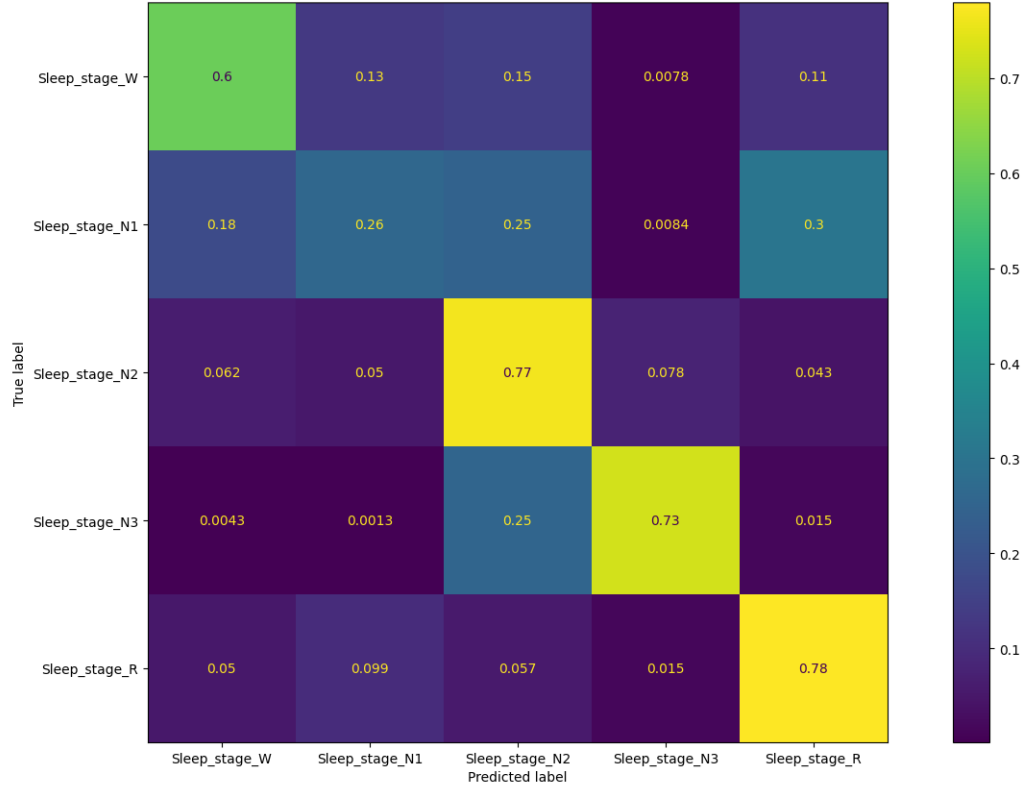


Figure 4: Confussion matrix

The confusion matrix clearly illustrates that the classifier struggles with the Sleep_stage_N1 class, which is frequently misclassified as either Sleep_stage_R or Sleep_stage_N2. This overlap is expected, as these stages share similar EEG characteristics and transitions between them can be subtle, especially in noisy or low-resolution data. In contrast, the remaining classes, Sleep_stage_N2, Sleep_stage_N3, and Sleep_stage_R, demonstrate significantly better performance, each achieving over 70% prediction accuracy. These results suggest that while the model is effective at distinguishing more clearly defined sleep stages, additional techniques are needed to improve its sensitivity to less distinctive stages like N1.

Data augmentation

Data augmentation or oversampling is a technique used to generate synthetic samples in order to balance class distributions within a dataset. These synthetic samples are typically created based on the existing instances of the target class, aiming to enhance the model's ability to learn from underrepresented data.

To address the issue identified in the previous section, specifically, the poor performance of the classifier on the Sleep_stage_N1 class, synthetic samples have been generated exclusively for this class. The goal is to increase its representation during training and assess whether this targeted oversampling can improve the network's ability to correctly classify Sleep_stage_N1 events, thereby reducing misclassification and improving overall model performance.

The confusion matrix obtained for this approach is the following:

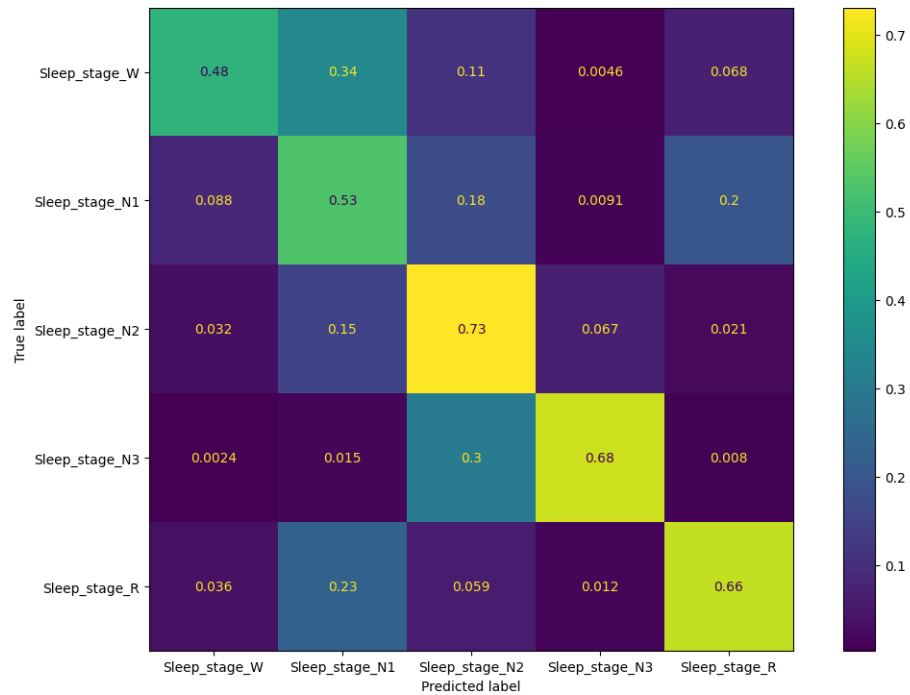


Figure 5: Confusion matrix oversampling N1 to equate the majority class

As shown in Figure 5, the classification performance for Sleep_stage_N1 improves following the application of oversampling; however, this enhancement comes at the expense of reduced accuracy in some of the other classes. While this trade-off is not ideal, it is worth noting that all classes except Sleep_stage_W maintain classification accuracies above 50%, indicating a generally acceptable level of performance across the majority of sleep stages.

In the following approach, a simple class balancing strategy was employed to address the issue of class imbalance observed in the dataset. Specifically, the mean number of samples across all sleep stage classes was computed to serve as a reference threshold. If the number of samples in Sleep_stage_N1 fell below this average, the class was oversampled, through synthetic data generation, until it reached the same number of instances as the calculated mean. This method aims to equalize class representation, thereby reducing model bias toward majority classes and improving the classifier's ability to learn from underrepresented sleep stages. The confusion matrix obtained is the following:

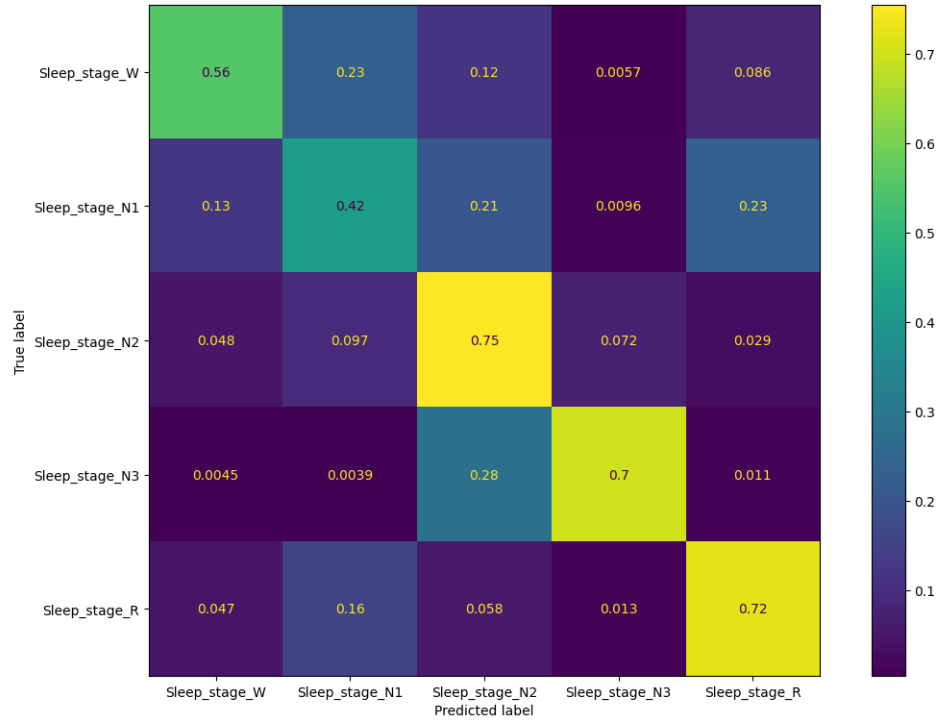


Figure 6: Confusion matrix oversampling N1 to achieve the mean

As shown in Figure 6, this approach yields results that closely resemble those obtained without any oversampling. However, it does lead to a notable improvement in the classification performance of Sleep_stage_N1, with its success rate increasing by over 15%. This gain, unfortunately, comes at the cost of reduced accuracy in other classes, highlighting a consistent trade-off observed across the experiments: oversampling a single class tends to negatively impact the classifiers performance on the remaining classes.

To address this limitation, a third approach was tested: oversampling all minority classes to match the sample size of the majority class. This method corresponds to the default configuration of the SMOTE (Synthetic Minority Over-sampling Technique) algorithm and aims to create a more balanced dataset overall, thereby reducing the model's bias and improving generalization across all classes. The confusion matrix obtained in this case is the following:

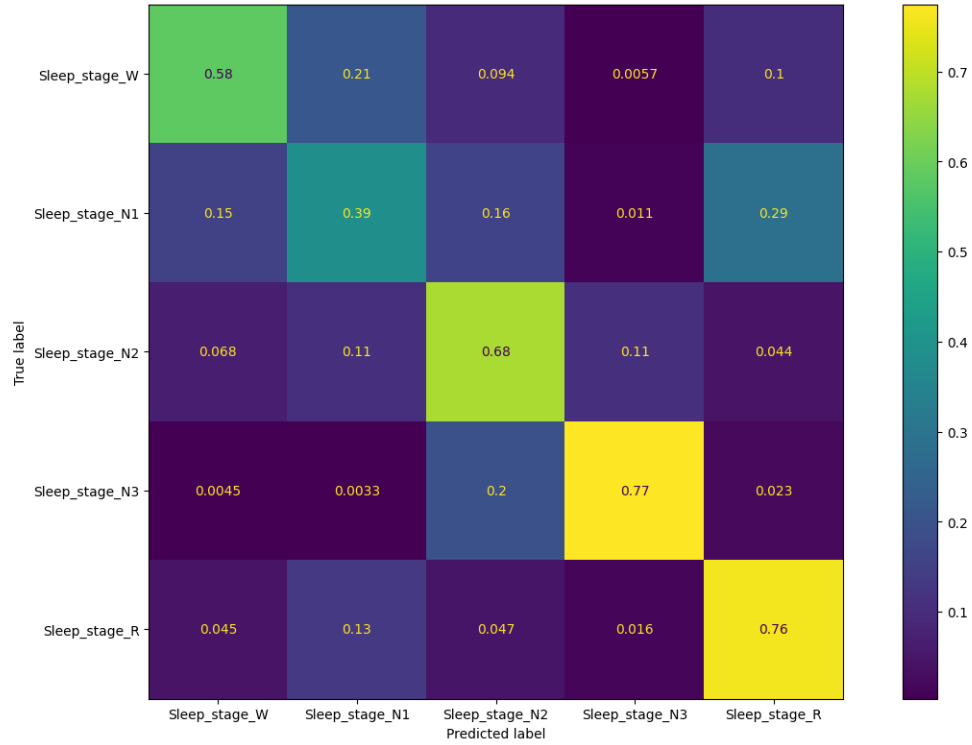


Figure 7: Confussion matrix oversampling all minority classes

As illustrated in Figure 7, this approach results in a reduction in overall classification accuracy. Moreover, the improvement in Sleep_stage_N1 detection is marginal, falling short of the initial objective of oversampling, which was to significantly enhance the model’s performance on this underrepresented class. These results suggest that while global oversampling may help balance the dataset, it does not necessarily translate into improved classification for the most challenging classes.

Since oversampling did not yield the expected improvements, an alternative strategy was considered: merging the classes Sleep_stage_N1 and Sleep_stage_N2 into a single cluster. While this approach sacrifices some granularity and precision in the classification results, it may lead to more robust and accurate predictions of the broader sleep stages. By reducing the number of classes, the model is expected to perform better in distinguishing between meaningful transitions in sleep, particularly in cases where the boundaries between

N1 and N2 stages are subtle and frequently confused, even by human experts.

Class clustering

By applying the class clustering technique, merging Sleep_stage_N1 and Sleep_stage_N2 into a single class, the model was trained on data from 1,371 subjects, each contributing an average of 850 samples, resulting in approximately 1.16 million samples in total. For evaluation, the same test set used in previous experiments was employed to ensure consistency and comparability of results. The classification report generated from this clustered model is presented below, providing insight into the impact of class merging on overall and per-class performance.

	precision	recall	f1-score	support
Sleep_stage_W	0.64	0.80	0.72	17240
Sleep_stage_N1/N2	0.83	0.73	0.78	58146
Sleep_stage_N3	0.66	0.76	0.70	13242
Sleep_stage_R	0.58	0.61	0.60	13843
accuracy			0.73	102471
macro avg	0.68	0.73	0.70	102471
weighted avg	0.74	0.73	0.73	102471

Table 5: Classification report

And the confusion matrix obtained is:

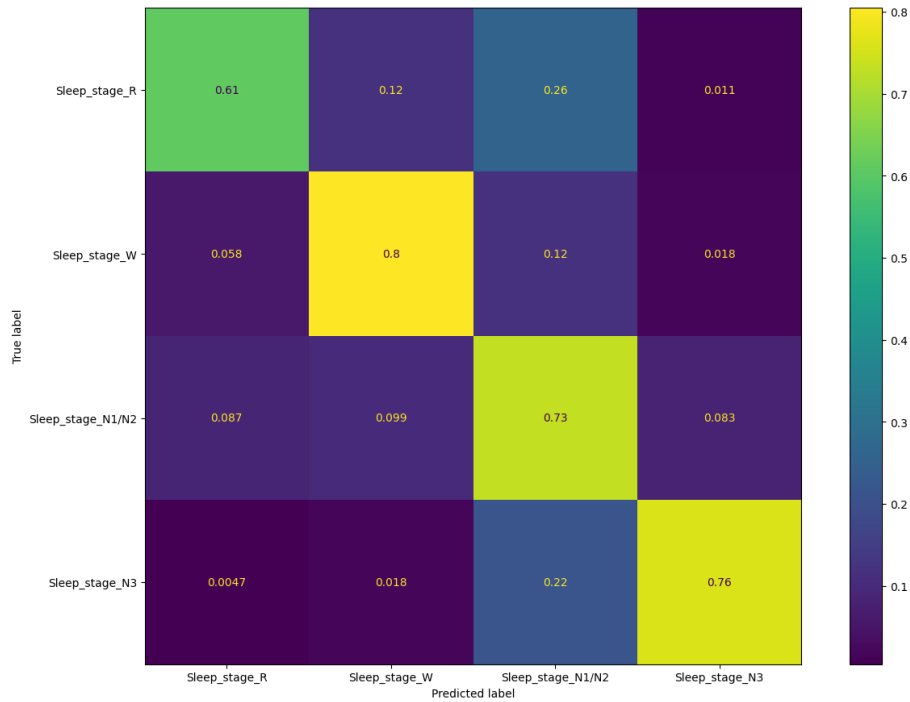


Figure 8: Confusion matrix clustering N1/N2

This time, the results are considerably more robust, enabling reliable classification of the different sleep stages. In particular, the model demonstrates strong capability in distinguishing between awake and asleep states. With this improved level of accuracy, we are now well-positioned to proceed with our primary objective: establishing meaningful connections between the Human Sleep Project dataset and the Soundless dataset.

Sleep Metrics

When investigating sensitivity to external stimuli during sleep, it is often beneficial to rely on well-established metrics to ensure consistency and comparability with existing research. One of the most widely used and informative metrics is sleep efficiency [8], which is defined as follows:

$$SE = TST/TIB$$

Where TST represents the total sleep time, and TIB denotes the total time spent in bed. Now that we are able to classify segments of electroencephalogram recordings into distinct sleep stages, we can compute the sleep efficiency for each subject. This metric enables us to quantitatively compare participants from the Soundless dataset, who sleep in real-world, potentially noisy environments, with subjects from the Human Sleep Project, whose recordings were obtained in highly controlled settings. By doing so, we aim to assess whether external noise significantly alters sleep quality for individuals with similar profiles.

Taking into account the different sleep phases of the subjects allows us to further refine our analysis. Deep sleep, also known as slow-wave sleep or stage N3, represents a critical phase of the sleep cycle due to its restorative functions. It is the deepest stage of non-REM sleep, characterized by slow brain waves (delta waves) and a marked reduction in physiological activity, including decreased heart rate, breathing rate, and muscle tone. This stage plays a vital role in physical recovery, immune system strengthening, memory consolidation, and hormone regulation.

In contrast, REM (Rapid Eye Movement) sleep is the final stage of the sleep cycle, distinguished by intense brain activity, vivid dreaming, and temporary muscle atonia (limb paralysis). REM sleep typically begins approximately 90 minutes after sleep onset and recurs in cycles throughout the night, with later cycles becoming progressively longer. This stage is widely believed to be essential for cognitive processes such as memory consolidation, learning, and emotional regulation.

By differentiating these stages in our data, we can better understand how external factors, such as ambient noise, may uniquely impact the restorative and cognitive functions associated with deep and REM sleep.

In addition to sleep efficiency, we will compute the percentage of the night that each subject spends in different sleep stages: light sleep (stages N1 and N2), deep sleep (stage N3), and REM sleep. These metrics will enable us to characterize the overall sleep architecture of each individual. Using these values, we plan to aggregate sessions from both the Soundless dataset and The Human Sleep Project to perform cluster analysis. This approach will help us identify distinct patterns or groupings in sleep behavior across different environments. Our analysis will encompass approximately 12 thousand sessions from The Human Sleep Project and 362 sessions from the Soundless dataset.

For the clustering analysis, we will employ the well-established KMeans algorithm. Prior to clustering, a Principal Component Analysis (PCA) is conducted to reduce the dimensionality of our variable space. Specifically, we reduce the four variables under study: sleep efficiency, percentage of time in light sleep, deep sleep, and REM sleep, into two principal components.

To determine the optimal number of clusters, we evaluate the clustering performance using the silhouette score across cluster counts ranging from 2 to 6. The analysis indicates that three clusters is the optimal choice, yielding a silhouette score of 0.392. The corresponding graphical representation is shown below:

Silhouette analysis for KMeans clustering on sample data with $n_clusters = 3$

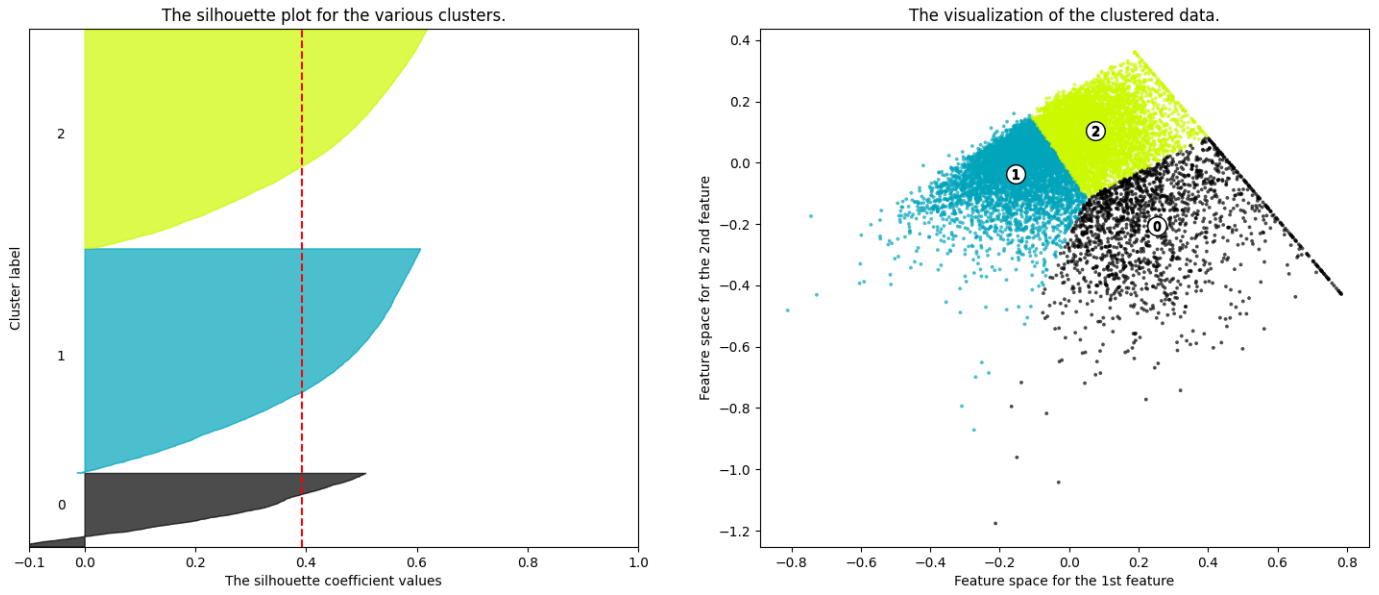


Figure 9: Silhouette score for 3 clusters

The graphics for 2, 4, 5 and 6 clusters are not included in this document for the sake of brevity.

With this information, we proceed with our study by performing KMeans clustering using the three identified clusters. This allows us to analyse and compare the sleep efficiency within each cluster. The results are summarized as follows:

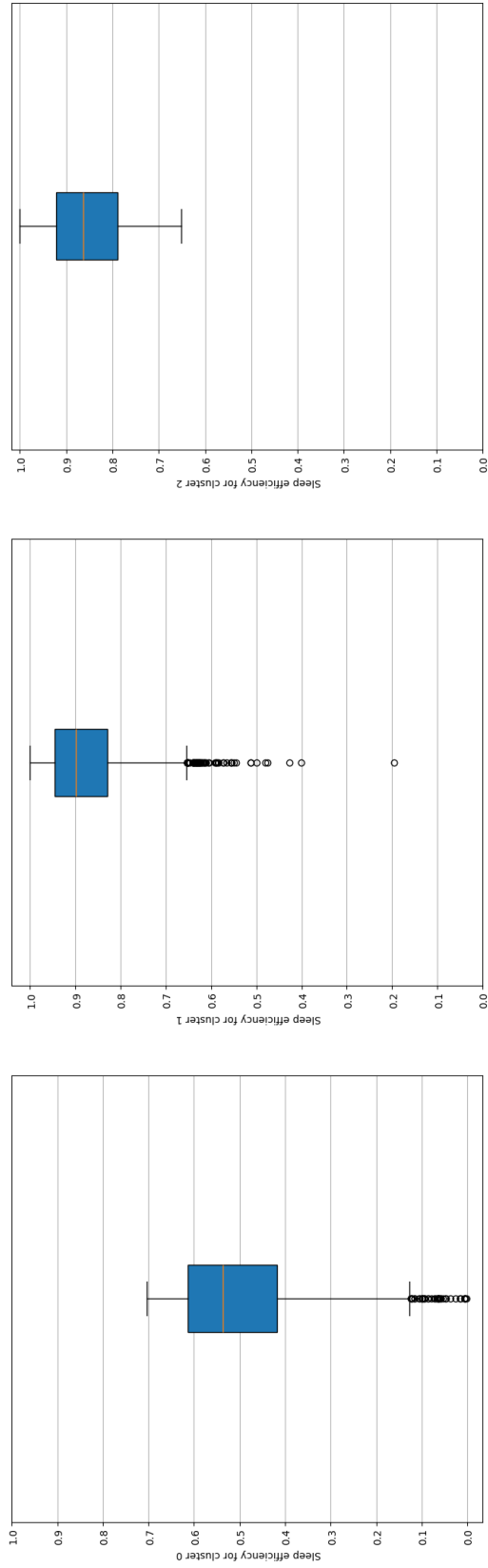


Figure 10: Sleep efficiency for 3 clusters

As shown in Figure 10, we can clearly distinguish three distinct user profiles based on sleep efficiency:

- Cluster 1 exhibits a low sleep efficiency, with a median around 0.54 and a lower quartile near 0.13. This suggests that subjects in this cluster experience significant sleep disturbances or fragmentation, pointing to severe difficulties maintaining sleep.
- Cluster 2 shows a median sleep efficiency of approximately 0.90, indicating generally good sleep quality. However, the presence of numerous outliers below the lower quartile (around 0.75) suggests that some subjects in this group still experience poor sleep episodes, potentially intermittently.
- Cluster 3 presents a median sleep efficiency around 0.87, also reflecting good sleep quality. Notably, this cluster contains no significant outliers, and its lower quartile of approximately 0.65 is relatively acceptable, indicating more consistent sleep patterns across subjects.

Regarding the distribution of Soundless sessions across clusters:

- 293 out of 362 sessions fall into Cluster 1,
- 55 sessions into Cluster 2, and
- only 14 sessions into Cluster 3.

This distribution strongly suggests that exposure to environmental noise, as captured in the Soundless dataset, negatively impacts sleep quality, with the majority of these subjects classified in the poorest sleep efficiency cluster.

We now delve further into the analysis by examining the distribution of sleep stages (light sleep, deep sleep, and REM) within each cluster to gain deeper insights into how sleep architecture is altered across the different user profiles:

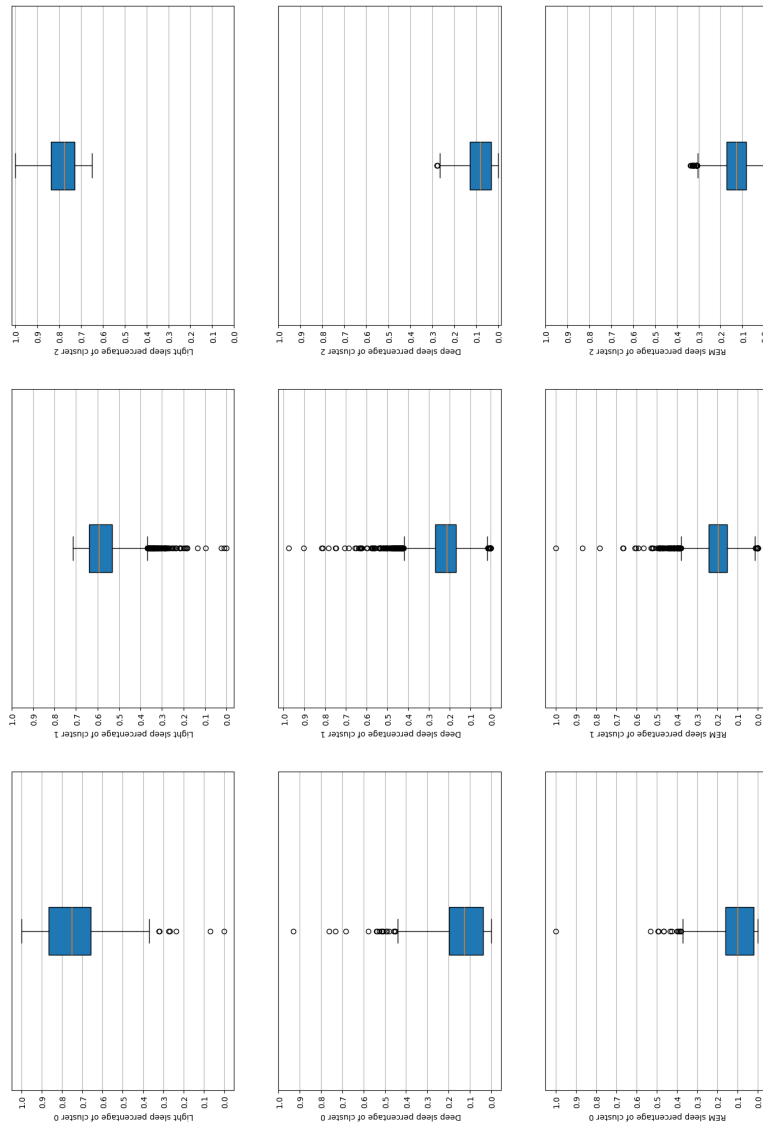


Figure 11: Sleep stages percentages for 3 clusters

The graphic in Figure 11 provides further insight into the distribution of sleep stages across clusters. For reference, in healthy adults:

- Light sleep, particularly Stage N2, typically comprises around 50% of total sleep time.
- Stage N1, the initial transition into sleep, is very brief.
- Deep sleep (Stage N3) generally ranges from 10% to 25% of the night.

- REM sleep accounts for approximately 20% to 25% of total sleep.

Subjects in the first cluster, previously identified as having low sleep efficiency, show a median light sleep percentage of around 75%, significantly above the expected range. This aligns with the earlier observation in Figure 10, where the same group demonstrated fragmented or non-restorative sleep.

In this cluster:

- Deep sleep has a median value around 12%.
- REM sleep is even lower, at approximately 10%.

These metrics suggest that individuals in the first cluster not only struggle to maintain efficient sleep but also have difficulty entering and sustaining deep or REM sleep, potentially due to environmental disturbances (e.g., noise) or underlying sleep disorders.

In the second cluster, subjects exhibit a median light sleep percentage of approximately 58%, much closer to typical values. The presence of numerous outliers below the lower quartile implies that many individuals in this group achieve even more restorative sleep.

Supporting this interpretation:

- Deep sleep median is around 22%.
- REM sleep median is close to 20%.

These figures align well with expected distributions in healthy sleep, indicating better sleep quality compared to the first cluster.

The third cluster appears to represent a transitional group with intermediate characteristics. In this group:

- Light sleep accounts for a median of 77%, which is unusually high.
- Deep sleep is only around 7%, significantly below normal.
- REM sleep is approximately 12%.

Such patterns may suggest the presence of underlying sleep pathologies or conditions that hinder restorative sleep. Supporting this interpretation is the fact that only 14 sessions from the Soundless dataset fall into this cluster, indicating that it may represent a more clinically affected subgroup within the broader population.

To further validate our findings, we analyse the environmental noise exposure by plotting the distribution of sound pressure levels (in decibels) recorded during the Soundless sessions, grouped by the clusters previously identified. This visualization enables us to

assess whether higher ambient noise levels correlate with poorer sleep efficiency and altered sleep stage distributions, thereby providing additional evidence for the hypothesis that environmental noise negatively impacts sleep quality.

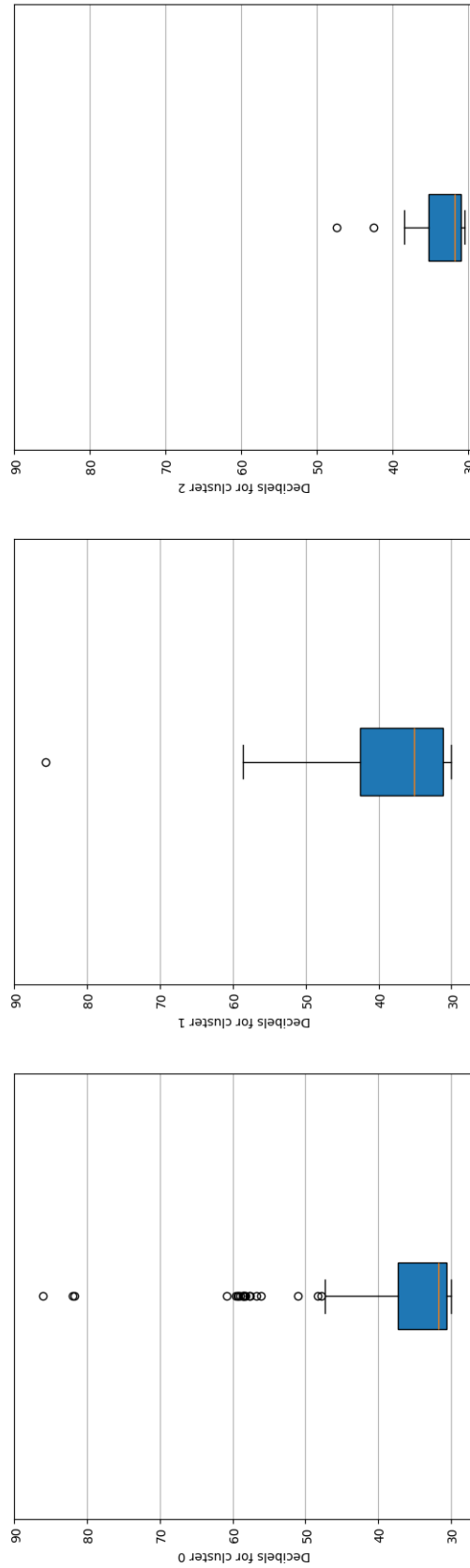


Figure 12: Decibels for soundless users for 3 clusters

In Figure 12, we observe that the first cluster, which, as previously discussed, includes the majority of individuals experiencing poor sleep quality, exhibits a median ambient noise level of approximately 32dB. While this median might suggest moderate noise exposure, the upper quartile exceeds 48dB, and numerous outliers surpass this threshold, indicating that a significant subset of these sessions involved exposure to elevated noise levels. Notably, 293 Soundless sessions fall into this cluster.

In the second cluster, which includes subjects that display a healthier sleep pattern, with consistent entry into deep and REM sleep stages, the median noise level is slightly higher at 33dB, but the distribution is more compact, with no significant outliers beyond the upper quartile of 38dB. Fifty-five Soundless sessions are included in this cluster.

In the third cluster, the smallest group with only 14 Soundless sessions, shows a median ambient noise level around 31dB, with an upper quartile near 39dB, indicating relatively low environmental noise exposure.

These findings support the hypothesis that ambient noise not only impacts overall sleep efficiency but also the physiological quality of sleep, particularly the ability to transition into and sustain deep sleep and REM stages. Subjects exposed to higher levels of environmental noise show sleep patterns that are qualitatively similar to those observed in individuals with diagnosed sleep pathologies.

In summary, the study presents the following conclusions:

- Individuals exposed to elevated night time noise levels tend to have reduced sleep efficiency.
- Noise exposure negatively affects the proportion of time spent in restorative sleep stages, such as deep sleep (N3) and REM.
- The impact of chronic noise exposure on sleep architecture can mimic the effects of clinical sleep disorders.

Conclusion

Sleep research is inherently complex, and introducing the variable of environmental noise adds an additional layer of intricacy. Individual variability plays a significant role in how noise affects sleep: some people are naturally more sensitive to sound, while others may remain undisturbed even under relatively noisy conditions. Factors such as age, stress levels, overall health, and personal noise tolerance contribute to the wide variability in sleep quality across individuals. Moreover, certain frequencies or sound patterns might disrupt one person's sleep while having no noticeable effect on another's.

Despite these challenges, this study has demonstrated that it is possible to model and predict sleep stages using EEG recordings through machine learning techniques, specifically a Multi-Layer Perceptron (MLP) architecture trained on power spectral density (PSD) features. This model provides a valuable framework for analysing how external factors, such as ambient noise, influence sleep architecture. These methods could also be adapted further to detect and classify arousals during sleep, enabling the correlation of noise events (recorded via environmental microphones) with disturbances in sleep stages.

The Human Sleep Project dataset, used extensively in this study, includes EEG data from subjects with known sleep pathologies. While this has allowed for robust testing of the proposed model, future studies would benefit from incorporating data from healthy individuals to better understand the full spectrum of sleep behaviour and to establish a baseline for what constitutes "normal" sleep unaffected by pathology or noise.

In a perfect experimental set-up, we would be able to design a controlled study where subjects are deliberately exposed to sounds of varying frequencies and amplitudes, allowing for precise measurement of thresholds at which sleep quality deteriorates. While such an ideal dataset is currently unavailable, the real-world data from the Human Sleep Project has allowed us to take a valuable first step in understanding how noise impacts sleep quality in more naturalistic conditions.

Turning to the Soundless dataset, which comprises data collected from real-world environments with varying noise levels, our findings are particularly compelling. The analysis revealed that noise is a significant factor influencing both sleep efficiency and sleep archi-

texture. Individuals exposed to higher levels of ambient noise, particularly above 45 dB, consistently displayed:

- Lower sleep efficiency (i.e., spending less of their time in bed actually sleeping),
- Reduced time spent in deep sleep (N3) and REM sleep, which are both critical for cognitive and physical restoration.

Out of the 362 Soundless sessions evaluated, 293 were categorized into the worst-performing cluster, marked by low sleep efficiency and minimal time spent in the restorative stages of sleep. In contrast, only 14 sessions fell into the best-performing cluster, where participants demonstrated healthy proportions of deep and REM sleep alongside higher sleep efficiency. This imbalance strongly suggests that noise exposure is not only prevalent but also detrimental to the quality of sleep for the majority of Soundless users.

From this, we can conclude that environmental noise is a determining factor in sleep degradation. Prolonged or frequent exposure to sound levels exceeding 45 dB appears sufficient to prevent individuals from achieving healthy sleep, both in quantity and quality. More alarmingly, the data indicates that individuals exposed to chronic night time noise may exhibit worse sleep patterns than those suffering from clinical sleep disorders, such as insomnia or chronic pain. In other words, noise pollution, often overlooked as a minor inconvenience, may pose a serious threat to public health by inducing sleep patterns consistent with medical pathology.

This study not only highlights the importance of protecting sleep environments from ambient noise but also opens the door to future research aimed at further quantifying the relationship between sound and sleep, potentially leading to the development of noise mitigation strategies, personalized sleep health tools, and policy recommendations for urban sound management.

Future work

As outlined in the conclusion, the methods developed in this study serve as a foundational step toward understanding how environmental noise influences sleep quality. The ability to classify sleep stages from EEG signals provides a powerful tool for identifying when, during the sleep cycle, noise has the most pronounced impact. Beyond sleep stage classification, a promising avenue for future exploration is the prediction of arousals (brief awakenings or disturbances during sleep) which can be correlated with external stimuli such as sound events.

One key area for future research is leveraging cross-subject generalization. In this study, we demonstrated that it is feasible to predict the sleep stage of a subject using a model trained on data from other individuals. Extending this approach to arousal prediction could yield valuable insights into how noise disrupts sleep patterns across a population. Detecting arousals from EEG signals and correlating them with concurrent noise events could provide a finer-grained understanding of sleep disruption.

Another direction for future work involves expanding and enriching the Soundless dataset. With a larger and more diverse dataset, it would be possible to perform more granular analyses of how different frequencies and intensities of noise affect sleep. For example, isolating power across frequency bands (e.g., low-frequency hums vs. high-pitched sounds) may help identify which types of noise are most disruptive. Additionally, investigating whether noise-induced disruptions are detectable in the physiological signals captured by wearable devices, such as heart rate variability, skin conductance, or motion, could open up non-invasive methods for sleep disturbance detection.

While EEG remains the gold standard for sleep analysis, integrating wearable sensor data from Soundless devices with EEG-derived information from datasets like The Human Sleep Project may offer a practical pathway to scalable sleep monitoring. Bridging these datasets could enable the training of multimodal models capable of predicting sleep quality or disruptions using only wearable data, a major advancement in real-world sleep research.

A particularly promising idea is to link user profiles across datasets. The Human Sleep

Project includes pre-sleep questionnaires in many cases, offering valuable metadata about the subject's health, lifestyle, and subjective sleep quality. Encouraging Soundless users to complete a similar form would enable a deeper comparative analysis between datasets. This could facilitate clustering subjects based not only on physiological data but also on personal characteristics and behaviours, potentially improving model performance and interpretability.

In summary, the future directions for this research include:

- Developing arousal detection models and studying their correlation with noise exposure.
- Expanding the Soundless dataset to include more sessions and a wider range of acoustic environments.
- Analysing frequency-specific noise effects on sleep using high-resolution audio data.
- Enhancing wearable signal analysis to detect subtle signs of sleep disturbance without EEG.
- Establishing user profiling through standardized questionnaires to bridge EEG and wearable datasets.
- Creating multimodal predictive models combining EEG and wearable data for robust sleep quality estimation.

Pursuing these directions will further the goal of building scalable, reliable systems to monitor and improve sleep health in everyday environments.

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